Industrial Microbiology	The study of exploitation of microbes for use in industrial processes. Examples include industrial fermentation and waste water treatment. This field also includes brewing, an important application of microbiology
Microbial Biotechnology	The study of manipulation of microorganisms at the genetic and molecular level to generate useful products
Food Microbiology and Dairy Microbiology	The study of microorganisms in food spoilage, foodborne illness and food production.

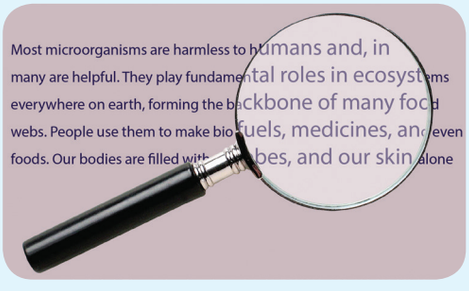
Summary

Microbiology is the study of microorganisms that includes bacteria, fungi,algae,protozoa and viruses. Many scientists contributed to the science of microbiology.

Antony Van Leuwenhoek made simple microscope. For the first time, Antony Van Leuwenhoek described the microorganisms. Louis Pasteur disproved the theory of spontaneous generation. Germ theory of disease came from the work of Pasteur and Robert Koch. Vaccines for Anthrax and rabies was developed by Pasteur. Direct relationship between the suspected pathogen and disease was established by Koch's postulates. Koch developed the technique of pure culture on solid medium. Joseph lister developed antiseptic surgery. Alexander Fleming discovered Penicillin. Waksman showed antimicrobian activity that led to the discovery of Streptomycin and other antibiotics. The branches of microbiology can be classified into pure and applied microbiology. Pure microbiology is classified based on taxonomical and integrated characteristics. Microbiology has got vast areas open for job opportunities.

Student Activity

1. Want to see spontaneous generation of life?
   Take chicken soup or meat soup boil it in a bottle. Keep it over the shadow of your window/or in a open place with mouth open. Observe for a week. You will see maggots (worms) growing. Observe and record your findings.
2. For you to enjoy-like Antony Van Leeuwenhoek !!
   Get a palmist lens, see through it a paper print. You will see letter becomes big, bigger, and at one point it is no longer magnifying the letter. A simple convex lens is magnifying things. Leeuweenhoek used such lens only. (as seen above) You know useful and useless magnification.



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